Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

For a monoatomic gas, the value of the ratio of $C_{p,m}$ and $C_{V,m}$ is

(A) $5/3$

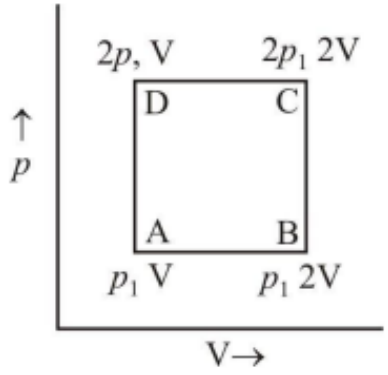
(B) $7/5$

(C) $9/7$

(D) $9/11$

Q.2

An ideal monatomic gas is taken round the cycle ABCD as shown in the fig. The work done during the cycle is



(A) pV

(B) $2pV$

(C) $\frac{1}{2}pV$

(D) zero

Q.3

The difference between heats of reaction at constant pressure and constant volume for the reaction $2\mathrm{C}_6\mathrm{H}_6(l)+15\mathrm{O}_2(\mathrm{~g}) \rightarrow 12\mathrm{CO}_2(\mathrm{~g})+6\mathrm{H}_2\mathrm{O}(l)$ at $25^{\circ}\mathrm{C}$ in kJmol^{-1} is

(A) -7.43

(B) $+3.72$

(C) -3.72

(D) $+7.43$

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Q.4

The standard enthalpies of formation at $300\ \mathrm{K}$ for $\mathrm{CCl_4}(l), \mathrm{H_2O(g)}, \mathrm{CO_2}(\ \mathrm{g})$ and $\mathrm{HCl(g)}$ are $-107, -242, -394$ and $-93 \mathrm{kJ mol^{-1}}$, respectively. The value of ΔU°_{300K} for the reaction $\mathrm{CCl_4}(l) + 2\mathrm{H_2O(g)} \rightarrow \mathrm{CO_2}(\ \mathrm{g}) + 4\mathrm{HCl(g)}$ is

(A) $-170\ \mathrm{kJ\ mol^{-1}}$

(B) $-175\ \mathrm{kJ\ mol^{-1}}$

(C) $-182.5\ \mathrm{kJ\ mol^{-1}}$

(D) $-282.5\ \mathrm{kJ\ mol^{-1}}$

Q.5

Which of the following reactions corresponds to the definition of enthalpy of formation at $298\ \mathrm{K}$?

(A) $\mathrm{H_2}(\ \mathrm{g}) + \frac{1}{2}\mathrm{O_2}(\ \mathrm{g}) \rightarrow \mathrm{H_2O}(l)$

(B) $2\mathrm{H_2}(\ \mathrm{g}) + \mathrm{O_2}(\ \mathrm{g}) \rightarrow 2\mathrm{H_2O}(l)$

(C) $\mathrm{H_2}(\ \mathrm{g}) + \frac{1}{2}\mathrm{O_2}(l) \rightarrow \mathrm{H_2O}(l)$

(D) $\mathrm{H_2}(\ \mathrm{g}) + \frac{1}{2}\mathrm{O_2}(\ \mathrm{g}) \rightarrow \mathrm{H_2O(g)}$

Q.6

For which one of the following equations is $\Delta_{\mathrm{r}}\mathrm{H}^{\circ}$ equal to $\Delta_{\mathrm{f}}\mathrm{H}^{\circ}$?

(A) $\mathrm{CH_4}(\ \mathrm{g}) + 2\mathrm{Cl_2}(\ \mathrm{g}) \rightarrow \mathrm{CH_2Cl_2}(l) + 2\mathrm{HCl(g)}$

(B) $\mathrm{Xe(g)} + 2\ \mathrm{F_2}(\ \mathrm{g}) \rightarrow \mathrm{XeF_4}(\ \mathrm{g})$

(C) $2\mathrm{CO(g)} + \mathrm{O_2}(\ \mathrm{g}) \rightarrow 2\mathrm{CO_2}(\ \mathrm{g})$

(D) $\mathrm{N_2}(\ \mathrm{g}) + \mathrm{O_3}(\ \mathrm{g}) \rightarrow \mathrm{N_2O_3}(\ \mathrm{g})$

Q.7

Internal energy of an ideal gas depends on

(A) Volume

(B) Temperature

(C) Pressure

(D) None of these

Q.8

The magnitude of enthalpy of neutralization of a weak acid and a strong base may be

(A) equal to $57.32\ \mathrm{kJ\ mol^{-1}}$

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(B) greater than $57.32\ \text{kJ mol}^{-1}$

(C) lesser than $57.32\ \text{kJ mol}^{-1}$

(D) greater or lesser than $57.32\ \text{kJ mol}^{-1}$ depending upon acid

Q.9

A well stoppered thermos flask contains some ice cubes. This is an example of

(A) Closed system

(B) Open system

(C) Isolated system

(D) Non-thermodynamic system

Q.10

$1\ \text{mol}$ each of $\text{CaC}_2, \text{Al}_4\text{C}_3, \text{Mg}_2\text{C}_3$ reacts with H_2O in a separate open flask. Numerical value of work done by the system is in order

(A) $\text{CaC}_2 < \text{Al}_4\text{C}_3 = \text{Mg}_2\text{C}_3$

(B) $\text{CaC}_2 < \text{Al}_4\text{C}_3 < \text{Mg}_2\text{C}_3$

(C) $\text{CaC}_2 = \text{Mg}_2\text{C}_3 < \text{Al}_4\text{C}_3$

(D) $\text{CaC}_2 = \text{Mg}_2\text{C}_3 = \text{Al}_4\text{C}_3$

Q.11

$8\ \text{g}$ of O_2 gas at 27°C is compressed to half of the volume under isothermal and reversible conditions. Work done on the system (in cal) is

(A) $\text{W} = 0$

(B) $\text{W} = 2.303 \times 150 \log 2$

(C) $\text{W} = -2.303 \times 150 \log 2$

(D) $\text{W} = -2.303 \times 300 \times 8 \times 2 \log 2$

Q.12

When one mole of an ideal gas is compressed to half its initial volume and simultaneously heated to twice its initial temperature, the change in entropy (ΔS) is

(A) $\text{C}_\text{V} \ln 2$

(B) $\text{C}_\text{p} \ln 2$

(C) $\text{R} \ln 2$

(D) $(\text{C}_\text{V} - \text{R}) \ln 2$

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Q.13

A reaction that is spontaneous can be described as:

(A) proceeding in both the forward and reverse directions

(B) having the same rate in both the forward and reverse directions

(C) releasing heat to the surroundings

(D) proceeding without external influence once it has begun

Q.14

The standard free energies of formation of $\mathrm{H_2\ S(\ g)}$ and $\mathrm{CdS(s)}$ at $1000^{\circ}\mathrm{C}$ are $-49.0\ \mathrm{kJ/mol}$ and $-127.2\ \mathrm{kJ/mol}$, respectively. Use these data to predict whether $\mathrm{H_2(\ g)}$ will reduce $\mathrm{CdS(s)}$ to metallic Cd at this temperature

(A) $\Delta \mathrm{G} = -78.2\ \mathrm{kJ/mol}$ and $\mathrm{H_2}$ reduces CdS

(B) $\Delta \mathrm{G} = -39.1\ \mathrm{kJ/mol}$ and $\mathrm{H_2}$ reduces CdS

(C) $\Delta \mathrm{G} = 0\ \mathrm{kJ/mol}$ and the reaction is at equilibrium

(D) $\Delta \mathrm{G} = +78.2\ \mathrm{kJ/mol}$ and the reaction is not feasible

Q.15

Identify the intensive quantity from the following

(A) Enthalpy and temperature

(B) Volume and temperature

(C) Enthalpy and volume

(D) Temperature and refractive index

Q.16

If a refrigerator's door is opened, then we get

(A) Room heated

(B) Room cooled

(C) More amount of heat is passed out

(D) No effect on room

Q.17

Molar heat capacity of water in equilibrium with ice at constant pressure is

(A) Zero

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(B) Infinity (∞)

(C) $40.45 \mathrm{~kJ~K^{-1}~mol^{-1}}$

(D) $75.48 \mathrm{~J~K^{-1}}$

Q.18

For the equilibrium $\mathrm{H_2O(l) \rightleftharpoons H_2O(g)}$ at $1 \mathrm{~atm}$ and $298 \mathrm{~K}$

(A) standard free energy change is equal to zero $(\triangle \mathrm{G^\circ} = 0)$

(B) free energy change is less than zero $(\Delta \mathrm{G} < 0)$

(C) standard free energy change is less than zero $(\triangle \mathrm{G^\circ} < 0)$

(D) standard free energy change is greater then zero $(\triangle \mathrm{G^\circ} > 0)$

Q.19

Which of the following statements is true

(A) $\Delta \mathrm{E}$ is always greater than $\Delta \mathrm{H}$

(B) $\Delta \mathrm{E}$ is always less than $\Delta \mathrm{H}$

(C) $\Delta \mathrm{E}$ may be lesser or greater or equal to $\Delta \mathrm{H}$

(D) $\Delta \mathrm{E}$ is always proportional to $\Delta \mathrm{H}$

Q.20

Which of the following is not a state function

(A) Internal energy

(B) Enthalpy

(C) Work

(D) Entropy

Q.21

The enthalpy of combustion of $\mathrm{H_2(~g)}$ to give $\mathrm{H_2O(g)}$ is $-249 \mathrm{~kJ~mol^{-1}}$ and bond enthalpies of $\mathrm{H-H}$ and $\mathrm{O=O}$ are $433 \mathrm{~kJ~mol^{-1}}$ and $492 \mathrm{~kJ~mol^{-1}}$, respectively. The bond enthalpy of $\mathrm{O-H}$ is (in $\mathrm{kJmol^{-1}}$)

Q.22

If the vaporization of benzene at its boiling point $(353 \mathrm{~K})$ is $31.064 \mathrm{~kJ~mol^{-1}}$, then its entropy of vaporization will be (in $\mathrm{JK^{-1}~mol^{-1}}$)

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Q.23

One litre-atmosphere is equal to ____ J(round off to the nearest integer)

Q.24

Temperature of one mole of helium gas is increased by $1^{\circ} \mathrm{C}$, hence increase in internal energy(in cal) is

Q.25

For an isomerisation reaction $\mathrm{A} \rightleftharpoons \mathrm{B}$, the temperature dependence of equilibrium constant is given by $\log_{\mathrm{e}} \mathrm{K}=4.0-\frac{2000}{\mathrm{T}}$
The value of $\Delta \mathrm{S}^{\circ} / \mathrm{R}$ at $300 \mathrm{~K}$ is therefore:

Q.26

How much energy (in kJ) must be supplied to change $36 \mathrm{~g}$ of ice at $0^{\circ} \mathrm{C}$ to water at room temperature, $25^{\circ} \mathrm{C}$

$\Delta \mathrm{H}^{\circ}{ }_{\text {fusion }}=6.01 \mathrm{~kJ} \mathrm{~mol}^{-1}$
$\mathrm{C}_{\text {p liquid }}=4.18 \mathrm{JK}^{-1} \mathrm{~g}^{-1}$

Q.27

The temperature of 4 moles of an ideal gas is raised from $300 \mathrm{~K}$ to $350 \mathrm{~K}$. What is the value of $\Delta \mathrm{H}-\Delta \mathrm{E}$ (in J) for this process?
$\left(\mathrm{R}=8.3 \mathrm{~J} \mathrm{~mol}^{-1} \mathrm{~K}^{-1}\right)$

Q.28

For the reaction $\mathrm{A}(\mathrm{s}) \rightleftharpoons \mathrm{B}(l)+2 \mathrm{C}(\mathrm{g}), \Delta \mathrm{G}^{\circ}$ (in joules) $=90800-100 \mathrm{~T}$.
The partial pressure of $\mathrm{C}(\mathrm{g})$ at $600 \mathrm{~K}$ in Torr is (round off to the nearest integer)

Q.29

What is the entropy change (in $\mathrm{JK}^{-1} \mathrm{~mol}^{-1}$) when one mole of ice is converted into water at $0^{\circ} \mathrm{C}$ (The enthalpy change for the conversion of ice to liquid water is $6.0 \mathrm{~kJ} \mathrm{~mol}^{-1}$ at $0^{\circ} \mathrm{C}$) (round off to the nearest integer)

Q.30

The enthalpy change $(\Delta \mathrm{H})$ for the process $\mathrm{N}_2 \mathrm{H}_4(\mathrm{~g}) \rightarrow 2 \mathrm{~N}(\mathrm{~g})+4 \mathrm{H}(\mathrm{g})$ is $1724 \mathrm{KJmol}^{-1}$. If the bond energy of the $\mathrm{N}-\mathrm{H}$ bond in ammonia is $391 \mathrm{KJmol}^{-1}$. What is the bond energy (in KJmol^{-1}) of $\mathrm{N}-\mathrm{N}$ bond in $\mathrm{N}_2 \mathrm{H}_4$ is

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Q.1 Answer:

$5/3$

Solution:

For a monoatomic gas, $\mathrm{C_{V,m}} = (3/2)\mathrm{R}$ and $\mathrm{C_{p,m}} = (5/2)\mathrm{R}$. Hence $\frac{C_{p,m}}{c_{V,m}} = \frac{5}{3}$

Q.2 Answer:

pV

Solution:

The work done during the cyclic path is equal to the area enclosed by the path, which is, pV.

Q.3 Answer:

-7.43

Solution:

$\Delta H - \Delta U = (\Delta V_{\mathrm{g}})\, RT = (-3)\left(8.314 \times 10^{-3}\ \mathrm{kJ\ K^{-1}\ mol^{-1}}\right)(298\ \mathrm{K}) = -7.43\ \mathrm{kJ\ mol^{-1}}$

Q.4 Answer:

$-182.5\ \mathrm{kJ\ mol^{-1}}$

Solution:

For the reaction:

$\mathrm{CCI_4}(l) + 2\mathrm{H_2O(g)} \rightarrow \mathrm{CO_2}(\mathrm{\ g}) + 4\mathrm{HCI(g)},$ we have

$\Delta_{\mathrm{r}}\mathrm{H} = \Delta_{\mathrm{f}}\mathrm{H}\left(\mathrm{CO_2},\ \mathrm{g}\right) + 4\Delta_{\mathrm{f}}\mathrm{H}(\mathrm{HCI}, \mathrm{g})$

$-\Delta_{\mathrm{f}}\mathrm{H}\left(\mathrm{CCI_4}, l\right) - 2\Delta_{\mathrm{f}}\mathrm{H}\left(\mathrm{H_2O}, \mathrm{g}\right)$

$= (-394 - 4 \times 93 + 107 + 2 \times 242)\mathrm{kJmol^{-1}} = -175\ \mathrm{kJ\ mol^{-1}}$

$\Delta_{\mathrm{r}}\mathrm{U} = \Delta_{\mathrm{r}}\mathrm{H} - (\Delta \mathrm{V_g})\,\mathrm{RT} = -175 - (3)$

$\left(8.314 \times 10^{-3}\ \mathrm{kJ\ K^{-1}\ mol^{-1}}\right)(300\ \mathrm{K}) =$

$-182.5\ \mathrm{kJ\ mol^{-1}}$

Q.5 Answer:

$\mathrm{H_2}(\mathrm{\ g}) + \frac{1}{2}\mathrm{O_2}(\mathrm{\ g}) \rightarrow \mathrm{H_2O}(l)$

Solution:

In the enthalpy of formation, we have a stoichiometric number of the desired species equal to one and it appears as the product. Moreover, the reactants must be in a stable state of aggregation.

Q.6 Answer:

$\mathrm{Xe(g)} + 2\ \mathrm{F_2}(\mathrm{\ g}) \rightarrow \mathrm{XeF_4}(\mathrm{\ g})$

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Solution:

The heat of formation is defined as the heat generated or absorbed when the compound is formed form its component elements in their standard state.
By this defination, 1 and 3 are clearly not correct as the starting materials are compounds.
4 is not correct, since ozone is not correct standard state of oxygen.
So, 2 is the correct answer.

Q.7 Answer:

Temperature

Solution:

In an ideal gas the inter-molecular collisions are assumed to be absent and the collisions are perfectly elastic. Thus, the gas possesses only translational kinetic energy and hence the internal energy of the ideal gas depends only on temperature.

Q.8 Answer:

lesser than $57.32\ \text{kJ mol}^{-1}$

Solution:

A portion of heat is utilized in the ionization of weak acid. Hence, the magnitude of enthalpy of neutralization is less than $57.32\ \text{kJ mol}^{-1}$

Q.9 Answer:

Isolated system

Solution:

An isolated system is a thermodynamic system that can not exchange either energy or matter outside the boundaries of the system.
Thus, a well stopped thermos flask containing ice is an example for an isolated system.

Q.10 Answer:

$\text{CaC}_2 = \text{Mg}_2\text{C}_3 < \text{Al}_4\text{C}_3$

Solution:

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$$CaC_{2(s)} + 2H_2O \rightarrow Ca(OH)_2 + C_2H_{2(g)}$$

1mol 1mol

$$Al_4C_{3(s)} + 12H_2O \rightarrow 4Al(OH)_3 + 3CH_{4(g)}$$

1mol 3mol

$$Mg_2C_{3(s)} + 4H_2O \rightarrow 2Mg(OH)_2 + C_3H_{4(g)}$$

1mol 1mol

Thus, $CaC_2 = Mg_2C_3 < Al_4C_3$

Thus, (c)

Q.11 Answer:

$\mathrm{W} = -2.303 \times 150 \log 2$

Solution:

work done $= 2.303\mathrm{nRT} \log \frac{\mathrm{V_2}}{\mathrm{V_1}} = 2.303 \times \frac{1}{4} \times 2 \times 300 \log \frac{1}{2} = -2.303 \times 150 \log 2\mathrm{cal}$

Thus, (c)

Q.12 Answer:

$(\mathrm{C_V} - \mathrm{R}) \ln 2$

Solution:

When there is simultaneously change in temperature and volume (or pressure)

$\Delta \mathrm{S} = nC_v \ln\left(\frac{T_2}{T_1}\right) + nR \ln\left(\frac{V_2}{V_1}\right)$

$= C_V \log_e\left(\frac{2}{1}\right) + R \log_e\left(\frac{1}{2}\right)$

$= C_V \log_e 2 - R \log_e 2$

$= (C_V - R) \log_e 2$

hence (d)

Q.13 Answer:

proceeding without external influence once it has begun

Solution:

A reaction that is spontaneous can be described as

(d) proceeding without external influence once it has begun. TRUE. A spontaneous reaction has a tendency to proceed in the way it has begun. It is spontaneous when Gibbs free energy is negative.

(c) releasing heat to the surroundings. FALSE. If the reaction is exothermic (enthalpy negative), one still has to take into account the effect of the entropy on Gibbs free energy.

(a) proceeding in both the forward and reverse directions. FALSE. This just means that the reaction is reversible.

(b) having the same rate in both the forward and reverse directions. FALSE. This is related to the

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kinetics but not the spontaneity.

So, the correct option is (d)

Q.14 Answer:

$\Delta G = +78.2\ kJ/mol$ and the reaction is not feasible

Solution:

$CdS + H_2 \rightarrow Cd + H_2S$

$\Delta G = \Delta G(H_2S) - \Delta G(CdS)$

$= -49 - (127.2)$

$= +78.2\ kJ$

Thus, (d)

Q.15 Answer:

Temperature and refractive index

Solution:

Thermodynamic quantities can be divided in two categories:

i) Intensive : An intensive property is a bulk property, meaning that it is a physical property of a system that does not depend on the system size or the amount of material in the system.

Examples of intensive properties include temperature, refractive index, density and hardness of an object.

ii) Extensive : An extensive property is a property that changes when the size of the sample changes.

Examples are mass, volume, length, and total charge.

In the given options, Temperature and refractive index are intensive quantities.

Q.16 Answer:

Room heated

Solution:

The average room temperature in the room will rise. The reason is that with the door open the temperature will start to rise inside the refrigerator. The thermostat will kick in and try to cool it back down. This means the motor is running, which means heat is being added to the room.

Q.17 Answer:

Infinity (∞)

Solution:

$C_P = \left(\frac{\delta H}{\delta T}\right)_P$, **At equilibrium** T **is constant i.e.** $\delta T = 0$

thus $C_P = \infty$

Q.18 Answer:

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free energy change is less than zero $(\Delta G < 0)$

Solution:

The process, $H_2O(l) \rightleftharpoons H_2O(g)$,
is an endothermic process, $(\triangle H = +ve)$ **and entropy increases during this change** $(\triangle S = +ve)$. **Hence this process is spontaneous at all temperatures above** $0^{\circ}C (T\triangle S > \Delta H$, **so** $\triangle G$ **is negative** $\triangle G = \Delta H - T\triangle S$ **). Thus free energy change** $(\triangle G)$ **will be less than zero (negative) at** 1 atm **and** 298 K.

Q.19 Answer:

ΔE may be lesser or greater or equal to ΔH

Solution:

ΔE **may be greater or lesser or equal to** ΔH
$\therefore \Delta H = \Delta E + P\Delta V = \Delta E + \Delta nRT$

Q.20 Answer:

Work

Solution:

Work is not a state function as during a process its value depends on the path followed. The value of enthalpy, internal energy and entropy depends on the state and not on the path followed to get that state, hence these are state functions.

Q.21 Answer:

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Solution:

For the equation

$$H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(g)$$
$$\Delta H = -249 \text{ kJ mol}^{-1}$$

Thus $\varepsilon(H-H) + \frac{1}{2}\varepsilon(O=O) - 2\varepsilon(O-H)$
$= -249 \text{ kJ mol}^{-1}$
$= \left[433 + \frac{1}{2}(492)\right] \text{kJmol}^{-1} - 2\varepsilon(O-H)$
$= -249 \text{ kJ mol}^{-1}$

Hence $\varepsilon(O-H) = \frac{1}{2}\left[+249 + 433 + \frac{1}{2}(492)\right] \text{kJmol}^{-1} = 464 \text{ kJ mol}^{-1}$

Q.22 Answer:

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Solution:

At the boiling point of benzene, $\Delta G = 0$. **Hence the expression** $\Delta G = \Delta H - T\Delta S$ **gives**

$\Delta S = \frac{\Delta H}{T} = \frac{31064 \text{ J mol}^{-1}}{353 \text{ K}} = 88 \text{ J K}^{-1} \text{ mol}^{-1}$

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Q.23 Answer:

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Solution:

$0.082\ \mathrm{L\ atm} = 8.314\ \mathrm{J}$. **Hence,** $1\ \mathrm{L\ atm}$
$= \frac{8.314}{0.082}\ \mathrm{J} = 101.3\ \mathrm{J}$

Q.24 Answer:

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Solution:

$C_v = \frac{\Delta E}{\Delta T}$
$\therefore \quad \Delta \mathrm{E} = \mathrm{C_V} \Delta \mathrm{T}$
$\mathrm{C_V} = \frac{3}{2}\mathrm{R}$ **for monoatomic gas**
$\Delta \mathrm{T} = 1^{\circ}$
$\therefore \quad \Delta \mathrm{E} = \frac{3}{2}\mathrm{R} = \frac{3}{2} \times 2 = 3\mathrm{cal}$

Q.25 Answer:

—

Solution:

Variation of K **with temperature is given by**
$\log \mathrm{K} = \frac{\Delta S^0}{R} - \frac{\Delta H^0}{2.303RT}$
$\log \mathrm{K} = 4.0 - \frac{2000}{T}$
$\therefore \quad \frac{\Delta S^0}{R} = 4$

Q.26 Answer:

—

Solution:

$\Delta \mathrm{H} = \Delta \mathrm{H}\,(36\ \mathrm{g}$ **ice at** $0^{\circ}\mathrm{C}$ **to** $36\ \mathrm{g H_2O}$ **at** $0^{\circ}\mathrm{C}) + \Delta \mathrm{H}\,(36\ \mathrm{g H_2O}$ **at** $0^{\circ}\mathrm{C}$ **to** $36\ \mathrm{g H_2O}$ **at** $25^{\circ}\mathrm{C})$
$= \Delta \mathrm{H^{\circ}_{fusion}} \times \mathrm{mol} + \mathrm{C_p} \times \mathrm{mass} \times \Delta \mathrm{T}$
$= 6.01 * \frac{36\ \mathrm{g}}{18g}\ \mathrm{kJ} + \frac{4.18}{1000} * 36 * 25\ \mathrm{kJ}$
$= 15.782 \approx 16\ \mathrm{kJ}$

Q.27 Answer:

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Solution:

$\mathrm{H} = \mathrm{E} + \mathrm{PV}$
$\mathrm{H} = \mathrm{E} + \mathrm{nRT}$
$\mathrm{H_2} = \mathrm{E_2} + \mathrm{nRT_2}$

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$H_1 = E_1 + nRT_1$

$\therefore \quad (H_2 - H_1) = (E_2 - E_1) + nR(T_2 - T_1)$

$\Delta H - \Delta E = nR\Delta T = 4 \times 8.3 \times 50$

$= 1660 \text{ J}$

Q.28 Answer:

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Solution:

A (solid) and B (liquid) does not contribute to partial pressure.

$K_p = p_c^2$

$\Delta G^\circ = 90800 - 100 \times 600$

$= 30800 \text{ J}$

$\Delta G = -2.303RT \log K_p$

$30800 = -2.303 \times 8.314 \times 600 \log p_c^2$

$\log_c^2 = -2.681$

$p_c = 0.0456 \text{ atm}$

$= 34.69 \text{Torr}$

Q.29 Answer:

—

Solution:

$dQ = TdS$

$\Rightarrow dS = \frac{dQ}{T}$

given that $dQ = \frac{6{\cdot}0 \times 10^3 \text{ Joule}}{\text{mole}}$ **at** 0^oC

& $T = 0^\circ C = 273K$

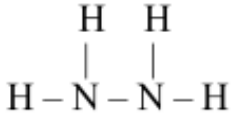
$\Rightarrow ds = \frac{6 \times 10^3}{273K} \ J/mole$

$= 21 \cdot 98 \frac{J}{\text{mole-K}}$

Q.30 Answer:

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Solution:



(So, $4 \text{ N} - \text{H}$ **bond present)**

means their energy $= 391 \times 4 = 1564$ **so the bond energy of** $\text{N} - \text{N}$ **in** $N_2H_4 = 1724 - 1564 = 160 \text{KJ/mol}$

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